

Customer Purchase Behavior Analysis

E-commerce case study

Data and analytical approach

RFM segmentation analysis to categorized customers by combining :

- **Recency** : Calculating last time order days
- **Frequency** : Calculating how many times customers had purchased
- **Monetary** : Calculating total spending



Analytical focus:

- Average orders days by segment
- Average order frequency by segment
- Repeat purchase by segment
- Product category by segment

Business problem / questions

How do customers behave based on the segmentation and what is the best strategy to treat each segmentation?

Champion customers contributed significant value to the business, yet **at risk** and **potential loyalist customers** plays critical role to maintain retention and frequency.....

At-risk customers have the **longest gap** between purchases, but still show relatively high order frequency. This suggests **declining engagement despite past activity**.

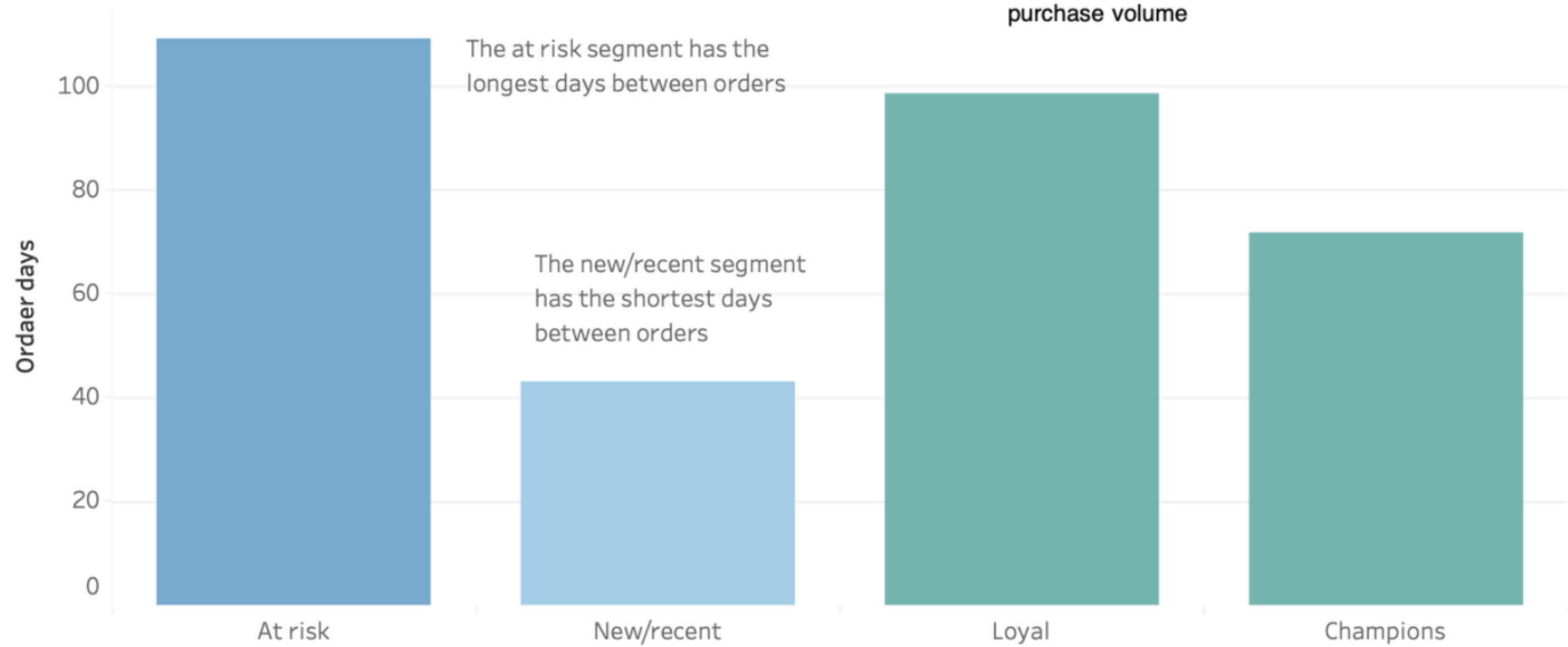
New/recent customers show the shortest purchase interval, indicating **strong initial engagement**.

Potential loyalist customers most likely doing **repeat purchase** but shows **low order frequency**.

Loyal customers generate the **second-highest revenue**, but are **not the strongest in purchase frequency** or consistency.

Champion customers contribute the **highest revenue and order frequency**, with **stable purchase intervals**, making them the **most valuable and consistent segment**.

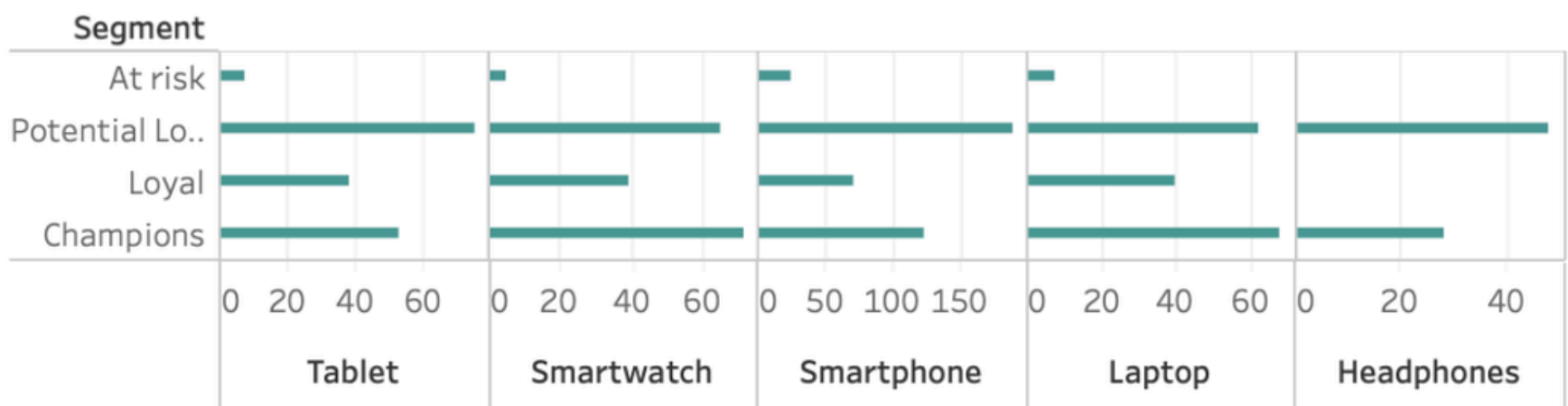
Average orders days by segment



Average order frequency by segment



Repeat purchase by segment



Product category by segment



Focus on **the high values customers** to maintain **the stability** and also focus on **the at-risk customers** to maintain **the retention.....**

At-risk customers should be **re-engaged through short-term incentives**, such as **limited-time discounts**, to encourage quicker repeat purchases and reduce the gap between transactions.

Potential loyalist customers can be encouraged to **increase purchase frequency** through **targeted incentives**, helping them transition into loyal customers.

Champion customers should be retained through **exclusive loyalty programs** or **membership benefits** to maintain their high value and consistency.